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From Radar Cardiography to Electrocardiograms: Conditional Diffusion Model Enabled Contactless ECG Monitoring Using mmWave Radar

Hanwen Zhang, Peichun Li, Liping Qian, *Senior Member, IEEE*, Zhiguo Shi, *Fellow, IEEE*, Yuan Wu, *Senior Member, IEEE*

***Abstract*—Cardiovascular disease (CVD) is one of the foremost causes of mortality globally, and cardiac arrhythmias constitute a major contributing factor. Continuous monitoring of cardiac signals plays a vital role for early detection and prevention. However, traditional electrocardiogram (ECG) devices require skin contact, which can be uncomfortable and inconvenient for long-term usage. In contrast, contactless cardiac health monitoring technologies, such as Wi-Fi and millimeter-wave (mmWave) radar, present a promising alternative. Millimeter-wave radar provides high range resolution, strong immunity to ambient light, and high sensitivity to small vibrations, making it ideal for contactless monitoring. However, mmWave radar primarily captures cardiac mechanical vibrations, known as radar cardiography (RCG) signals, which differ from the electrical activity recorded by clinical ECGs. To bridge this gap, we propose a contactless framework that uses mmWave radar and a conditional diffusion model to reconstruct ECG signals and then utilizes a deep learning model to classify arrhythmias. Specifically, mmWave radar captures RCG signals associated with cardiac activities. Leveraging the nonlinear relationship between cardiac mechanics and electrical activities, we design a Residual Network (ResNet)-based conditional diffusion model to convert these RCG signals into ECG signals. Finally, we develop a CNN-BiLSTM-SE network for arrhythmia classification. Experimental findings demonstrate the efficacy of the proposed approach for signal conversion as well as arrhythmia classification, offering a promising pathway toward contactless cardiac health monitoring.**

***Index Terms*—Contactless monitoring, mmWave radar, RCG and ECG signals, generative AI, arrhythmia classification.**

I. Introduction

In 2022, cardiovascular disease (CVD) accounted for an estimated 19.8 million deaths worldwide, making it one of the predominant causes of mortality, and this number continues to rise annually [1]. Arrhythmias are the major contributor to CVD-related mortality [2]. Therefore, continuous monitoring of cardiac signals is crucial for early arrhythmia detection and prevention. In clinical practice, cardiac activity is typically monitored using electrocardiogram (ECG) signals. However, traditional ECG devices (such as Holter monitors and electrocardiographs) require multiple electrodes attached to the skin. This contact-based method can cause irritation or discomfort and is particularly unsuitable for infants or patients with severe burns [3]. These limitations have motivated the exploration of contactless cardiac monitoring technologies.

Currently, the most widely used contactless cardiac monitoring technologies include video-based photoplethysmography (PPG) [4], Wi-Fi-based methods [5], and millimeter-wave (mmWave) radar [6]. Among these methods, video-based techniques are cost-effective. However their performance is limited by cosmetics, skin tone, and lighting conditions. Wi-Fi-based methods are widely accessible, yet most commercial systems lack sufficient bandwidth to resolve multipath interference along the range dimension. In contrast, mmWave radar has gained popularity in contactless cardiac monitoring due to its high range resolution, robustness to ambient light, low privacy risk, and high sensitivity to subtle vibrations.

Although these contactless methods can reliably estimate heart rate, they primarily sense chest-wall mechanical vibrations, known as radar cardiography (RCG) signals, which differ from the ECG signals used for arrhythmia diagnosis. ECG signals contain distinctive patterns, including the P wave, QRS complex, and T wave, which have well-defined clinical significance and serve as a critical basis for diagnosis [7]. Therefore, a key challenge is to convert RCG signals acquired through contactless monitoring into ECG signals.

From a physiological perspective, there is a well-established correlation between the heart's electrical and mechanical activity [8], [9]. Electrical signals in the myocardium trigger mechanical contractions, producing rhythmic motion. This coupling provides the physiological basis for converting RCG signals into their ECG counterparts. However, traditional signal processing methods based on physical models struggle to capture this highly nonlinear, time-dependent transformation. Recently, the rise of generative artificial intelligence (AI), especially diffusion models, has provided a new approach to complex signal conversion [10], [11]. Diffusion models learn the underlying data distribution by progressively adding controlled noise to low-noise signals and then reconstructing the target through a reverse denoising process [12], [13]. Owing to their training stability and ability to produce high-

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H. Zhang, P. Li, and Y. Wu are with the State Key Laboratory of Internet of Things for Smart City, University of Macau, Macau SAR, China, and also with the Department of Computer and Information Science, University of Macau, Macau SAR, China (e-mail: hw.zhang@connect.um.edu.mo, peichun.li@connect.um.edu.mo, yuanwu@um.edu.mo).

L. Qian is with the School of Ocean Engineering and Technology, Sun Yat-Sen University, Zhuhai 519082, China, and also with the Institute of Cyberspace Security, Zhejiang University of Technology, Hangzhou 310023, China (email: qianlp@mail.sysu.edu.cn).

Z. Shi is with the College of Information Science and Electronic Engineering, Zhejiang University, Hangzhou 310027, China (email: shizg@zju.edu.cn).

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quality outputs, diffusion models have been widely adopted in various signal generation scenarios [14].

In addition, deep learning has been widely applied to medical diagnosis in recent years since its powerful ability in feature representation and pattern recognition, effectively reducing the dependence on clinician experience [15]. In particular, recurrent neural networks, such as long short-term memory (LSTM), and convolutional neural networks (CNNs) can accurately capture the signals' spatial and temporal characteristics, improving diagnostic efficiency and accuracy.

Building on this background, this paper introduces a contactless ECG monitoring method that integrates mmWave radar and a conditional diffusion model. First, we propose a contactless sensing method that utilizes mmWave radar to detect and capture RCG signals associated with the human heartbeat. Next, we summarize key ECG characteristics and design a Residual Network (ResNet)-based conditional diffusion model incorporating an attention mechanism to convert RCG signals into high-quality ECG signals. Subsequently, we develop a deep learning framework that combines CNNs, bidirectional long short-term memory (BiLSTM) networks, and squeeze-and-excitation (SE) channel attention to classify cardiac arrhythmias. Finally, we assess the proposed method through experiments, confirming its utility in signal conversion and arrhythmia classification.

The main contributions of this work can be summarized as follows:

- We introduce a contactless ECG monitoring framework that utilizes mmWave radar to localize the target and capture RCG signals associated with the heartbeat, which are then converted into ECG signals for health monitoring.
- We analyze the characteristics of ECG signals and design a ResNet-based conditional diffusion model for signal conversion. The model effectively converts RCG signals into ECG signals, enhancing the practicality of contactless monitoring in clinical settings.
- We design a hybrid CNN-BiLSTM-SE neural network that effectively models both the spatial morphology and temporal dynamics of ECG signals while focusing on the most informative parts, leading to more reliable and accurate arrhythmia classification.
- Through experiments, we verify the effectiveness and advantages of the proposed framework, validating the feasibility of contactless cardiac health monitoring.

The remainder of this paper is structured as follows. Section II reviews related work on contactless cardiac monitoring with mmWave radar, generative AI-driven RCG to ECG signal conversion, and deep learning-based arrhythmia classification. Section III details the workflow of the proposed framework and explains how mmWave radar is used to acquire RCG signals induced by cardiac motion. Section IV reviews the characteristics of ECG signals and presents the proposed ResNet-based conditional diffusion model and CNN-BiLSTM-SE neural network in detail. In Section V, we present the experimental setup and results. Finally, Section VI concludes the paper and highlights the directions for future research.

II. Related Work

A. Contactless cardiac monitoring with mmWave radar

Most prior work on cardiac monitoring has focused on heart rate estimation, emphasizing signal quality enhancement and robust separation of heart rate components. These approaches generally fall into two categories: physics-based signal processing methods [16], [17] and data-driven deep learning methods [18], [19]. Physics-based approaches leverage mmWave signal propagation characteristics to perform beamforming, phase demodulation, harmonic suppression, etc. In previous work [16], the authors proposed a hierarchical enhancement strategy for wireless sensing and a cumulative positioning enhancement method, both designed to enhance the signal-to-noise ratio (SNR) of vital signs. Reference [17] derives a Newton-based maximum likelihood estimator to improve the estimation of vital sign parameters, thereby enabling more accurate heart rate assessment. In contrast, data-driven deep learning trains neural networks on large, well-labeled radar datasets to enhance vital sign monitoring in complex environments or when data are partially missing. HeRe [18] utilizes neural networks to reconstruct RCG signals, identify patterns in radar measurements, and thereby improve the SNR of the RCG signals. In another task [19], ResNet-18 identified time intervals of stable movement, and variational mode decomposition (VMD) then extracted the RCG signal from those intervals.

The methods above all aim to mitigate multipath interference in complex environments, emphasizing robust signal separation. However, the resulting RCG signals reflect cardiac mechanics rather than the ECG activity used for diagnosis.

B. Cardiac RCG to ECG Signal Conversion

Research on converting contactless RCG signals into ECG signals has grown rapidly in recent years. An early study [9] introduced an end-to-end model that employs a hybrid CNN–Transformer encoder together with a temporal convolutional network (TCN) decoder, mapping RCG signals to ECG signals based on excitation–contraction coupling and body-surface conduction. Building on ResNet's effectiveness for signal conversion, ResNet-based encoder–decoder architectures for RCG signals translation have been developed, e.g., RSSR-Net [20], and radarODE [21]. With the rise of generative AI, Generative Adversarial Networks (GANs)-based approaches have also emerged; representative examples include WGAN [22] and Attention-DCGAN [23]. However, encoder–decoder methods often miss details or oversmooth peaks in generated ECG signals, while GAN-based methods are vulnerable to mode collapse.

Diffusion models, in contrast, better preserve high-frequency components and produce more controllable outputs. AirECG [24] conducted an initial exploration of RCG to ECG signal conversion using a cross-domain diffusion model. However, AirECG mainly uses the cross-domain diffusion model built on a hierarchical Transformer with adaptive normalization. In contrast, our diffusion network adopts ResNets with time-step embeddings and SE attention, and applies global self-attention only once at the bottleneck. This design

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TABLE I
SUMMARY OF NOTATIONS

Symbols	Descriptions
$R(t)$, $d(t)$	Heartbeat-induced chest displacement (RCG signal), instantaneous radar–chest distance at time t.
d', c, λ	Nominal radar–chest distance, speed of light, electromagnetic wavelength.
A_{tx}, f_{tx}, B, T_{tx}	Transmit signal amplitude, starting frequency, bandwidth, chirp duration.
$s_{tx}(t)$, $s_{rx}(t)$, $s_{IF}(t)$	Transmitted FMCW signal, reflected echo signal, IF signal at time t.
$f_{IF}(t)$, $\varphi_{IF}(t)$	Instantaneous frequency and phase of IF signal.
N, S_E, $\{\beta_n\}_{n=1}^{N}$	Total number of diffusion steps, number of samples in each ECG segment, predefined variance schedule.
$\mathbf{R}$, $\mathbf{E}_{\text{gen}}$	RCG segment used as conditional input, generated ECG segment after the reverse diffusion process.
$\mathbf{E}_0$, $\mathbf{E}_n$, $\mathbf{E}_N$	Initial low noise ECG segment, noisy ECG at time n, pure Gaussian noise segment.
$q(\cdot)$, $\mathcal{N}(\cdot; \upsilon, \zeta)$, $\mathbf{I}$	Conditional probability distribution, multivariate Gaussian distribution with mean υ and covariance ζ, identity matrix.
ε, S_R, θ	Gaussian noise sample, number of samples in each RCG segment, set of all trainable parameters in the reverse network.
$p_\theta(\cdot)$, σ_n^2, μ_θ	Learnable reverse transition distribution, variance of the reverse process, mean value.
ε_θ, ε_r, $\mathcal{L}(\theta)$	Predicted noise, Gaussian noise injected during the reverse process, mean squared error during model training.
$\mathbf{h}$, $\mathbf{h}'$, $F_\varpi(\cdot)$	Residual block's input feature map, output feature map, learnable residual mapping.
ϖ, $\mathbf{h}^{(n)}$, $\mathbf{h}^{(n,R)}$	Set of trainable parameters of residual block, intermediate representation between $\mathbf{h}$ and $\mathbf{h}'$, $\mathbf{h}^{(n)}$ and $\mathbf{h}'$.
$\mathbf{e}_n$, $\mathbf{g}_R$, s_{ch}	Time embedding of step n, RCG feature embedding representation, channel-wise descriptor after global average pooling.
C^{se}, S^{se}, $u_{ch,\ell}$	Number of channels, time samples of the SE module input, value at channel ch and time index ℓ.
z_{ch}, $\nu(\cdot)$, $u'_{ch,\ell}$	gating coefficient for channel ch, sigmoid function, reweighted feature value after SE attention.

better matches the local waveform priors of ECG signals and reduces computational complexity, enabling stable training and efficient inference on limited clinical data.

C. Deep learning-based arrhythmia classification

Deep learning addresses tasks by learning patterns from data. Owing to its powerful capacity to capture high-dimensional and nonlinear relationships, it has been widely applied across medical domains [25], [26]. Notably, recent deep learning–based studies on arrhythmia classification report that detection accuracies have exceeded the average performance of cardiologists [27], [28]. Therefore, further improvement in the accuracy of arrhythmia classification has significant clinical implications. Researchers have explored single classifiers (e.g., CNN [27], BiLSTM [29]) and combined architectures (e.g., CNN–BiLSTM [30]) for arrhythmia classification.

This work develops a hybrid CNN-BiLSTM-SE network that captures spatial and temporal ECG features while emphasizing the most informative components.

III. System Overview and mmWave Radar–Based RCG Signals Acquisition

In this section, we first present the proposed system workflow and then illustrate how mmWave radar, in particular frequency-modulated continuous-wave (FMCW) radar, senses heartbeat-induced chest motion and extracts the RCG signals used for subsequent conversion. Table I summarizes the key notations used in this work.

A. System Workflow

We consider a single-target scenario for RCG signals monitoring using mmWave radar. The system provides a complete pipeline from contactless sensing to arrhythmia classification, as shown in Fig. 1. The workflow comprises three main stages:

- **Step 1. Contactless sensing**: Using mmWave radar to localize and track the chest wall of the target, remove static clutter, unwrap phase, and apply band-pass filtering to isolate the heartbeat component, obtaining the RCG signals.
- **Step 2. Signal conversion**: Treating the RCG signal as conditional input, a ResNet-based conditional diffusion model reconstructs the corresponding ECG segment through a denoising process.
- **Step 3. Arrhythmia classification**: The reconstructed ECG is processed by a deep network combining CNN, BiLSTM, and SE channel attention, which outputs multi-class arrhythmia labels.

The entire process operates under contactless conditions. In the following sections, we provide detailed explanations for each step.

B. Contactless Sensing Using mmWave Radar

1) Signal Model for Heartbeat-Induced Phase Modulation: During contactless sensing, our goal is to obtain the heartbeat-induced chest displacement at time t, i.e., the RCG signal $R(t)$. We assume that the target's chest is at the same height as the radar and that the radar beam points directly at the chest. Let d' denote the nominal radar–chest distance. The

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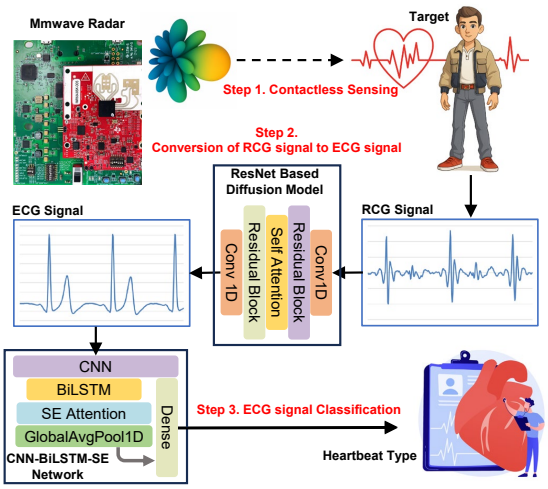

Fig. 1. Overview of the proposed system workflow.

instantaneous distance $d(t)$ is then the sum of d' and the heartbeat-induced displacement, i.e.,

$$d(t) = d' + R(t). \tag{1}$$

For FMCW radar, the transmitted signal $s_{tx}(t)$ [3], [31] is given by

$$s_{tx}(t) = A_{tx} \cdot \exp\{j2\pi(f_{tx}t + \frac{B}{2T_{tx}}t^2 + \varphi_0)\}, \tag{2}$$

where A_{tx}, f_{tx}, B, and T_{tx} denote the transmit signal amplitude, starting frequency, bandwidth, and chirp duration, respectively. The constant φ_0 denotes the initial phase.

Assuming that the reflection coefficient of the target is ρ, the reflected signal $s_{rx}(t)$ received by the radar is

$$s_{rx}(t) = \rho A_{tx} \cdot \exp\{j2\pi(f_{tx}(t - \frac{2d(t)}{c}) + \frac{B}{2T_{tx}}(t - \frac{2d(t)}{c})^2 + \varphi_0)\}, \tag{3}$$

where c represents the light speed.

Through coherent demodulation with a local oscillator, the transmitted and received signals are mixed to generate an intermediate-frequency (IF) signal [16]. Under the range correlation effect, the constant phase term φ_0 and the higher-order term $\frac{2Bd(t)^2}{T_{tx}c^2}$ can be neglected [32], yielding the approximate IF signal $s_{IF}(t)$:

$$s_{IF}(t) = s_{rx}(t) \cdot s_{tx}^{\star}(t) \approx \rho A_{\text{tx}}^2 \cdot \exp\{j2\pi(\frac{2Bd(t)}{cT_{tx}}t + \frac{2d(t)}{\lambda})\}, \tag{4}$$

where λ is the electromagnetic wavelength.

2) RCG Signal Extraction: According to Eq. (4), the instantaneous frequency $f_{IF}(t)$ and phase $\varphi_{IF}(t)$ of the IF signal are

$$f_{IF}(t) = \frac{2B\big(d' + R(t)\big)}{cT_{tx}}, \quad \varphi_{IF}(t) = \frac{4\pi\big(d' + R(t)\big)}{\lambda}. \tag{5}$$

Since the nominal radar–chest distance d' is constant, we can derive the RCG signal as

$$R(t) = \frac{\lambda}{4\pi}\varphi_{IF}(t) - d'. \tag{6}$$

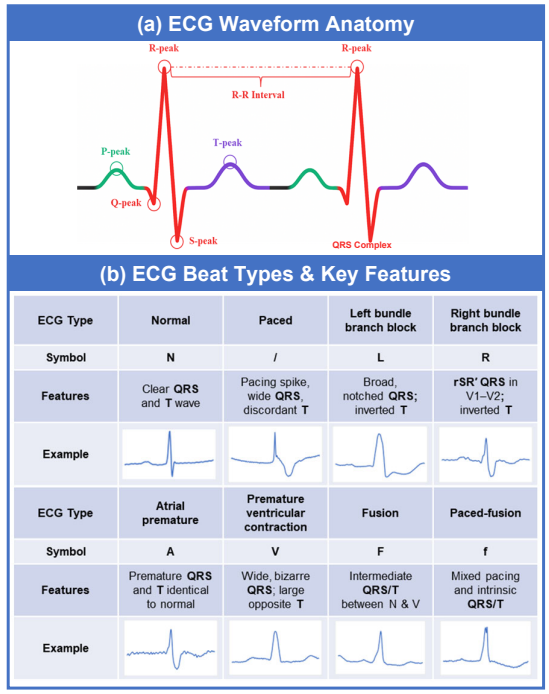

Fig. 2. **(a)**: Typical ECG morphology with annotated events. **(b)**: Representative ECG rhythm types and their key characteristics.

Accordingly, the RCG signal $R(t)$ can be recovered by tracking the phase of the IF signal and then used as the conditioning signal in the reverse process of the conditional diffusion model, which is described in Section IV-B. In practical measurements, the extracted phase is influenced by body motion, environmental noise, and respiratory harmonics. These effects can be mitigated either through preprocessing (e.g., filtering and artifact removal) or by employing RIS-assisted sensing to improve signal quality and suppress interference [33].

IV. Converting RCG to ECG

This section first reviews the fundamental morphology of ECG signals, the arrhythmia types considered in this work, and the principles of the conditional diffusion model. It then presents our proposed conditional diffusion–based ECG generation model and the CNN-BiLSTM-SE neural network.

A. Overview of ECG and Arrhythmia

The ECG represents cardiac electrical activity during each heartbeat. As illustrated in Fig. 2(a), a typical waveform comprises the P wave, the sharp QRS complex, and the T wave [34]. The P wave reflects atrial depolarization; the QRS complex corresponds to rapid depolarization of the left and right ventricles; the T wave represents ventricular repolarization; and the R–R interval indicates heart rate. In our setting, accurately reconstructing these ECG structures from the RCG is crucial, as they carry clinically meaningful cues for downstream arrhythmia classification.

An arrhythmia is any disturbance in impulse formation or conduction that disrupts normal cardiac rhythm and can alter the morphology or duration of the P wave, QRS complex, T

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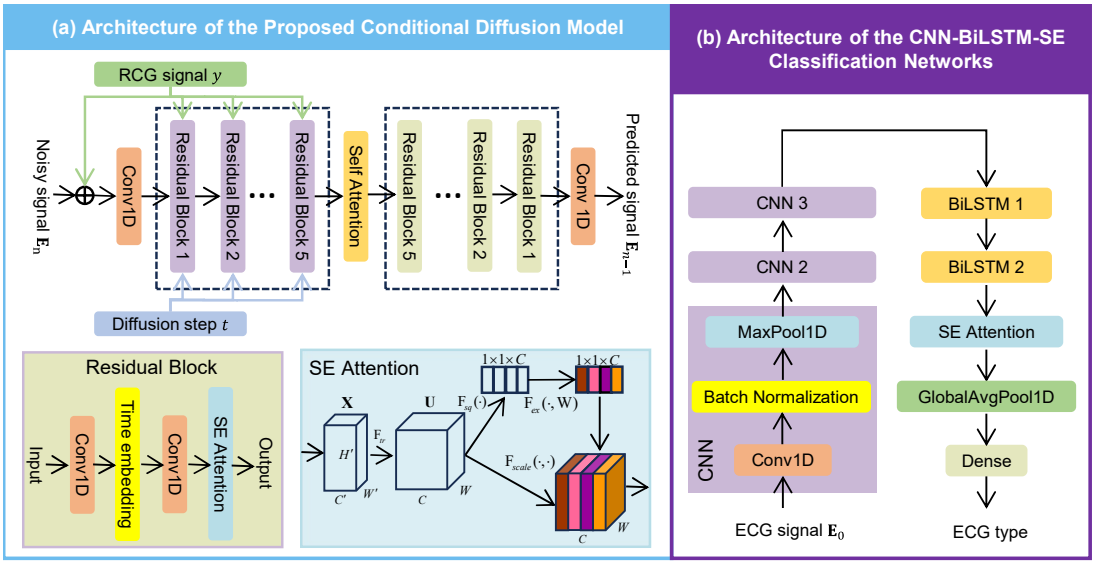

Fig. 3. Architectures of the proposed diffusion-based ECG synthesis and arrhythmia classification framework. (a) Conditional diffusion model denoising noisy ECG segments conditioned on RCG signals. (b) CNN-BiLSTM-SE classifier using ECG segments to predict heartbeat types.

wave, and related intervals [35]. Arrhythmias include many subtypes. In this work, we select seven representative arrhythmia types and the normal beat type from the MIT-BIH dataset [36] as the targets for subsequent classification. The characteristics and representative morphologies of these eight rhythm types are summarized in Fig. 2(b). From the above description, the morphology and duration of the P wave, QRS complex, and T wave emerge as key features for identifying abnormal heart rhythms.

B. Overview of the Conditional Diffusion Model

The conditional diffusion model generates data by learning to reverse a gradual noising process [37]. During training, low-noise ECG segments are progressively corrupted with Gaussian noise. During inference, the model is initialized with pure noise and iteratively denoises it to recover an ECG waveform. Compared with GANs, diffusion models provide more stable training and better mode coverage, properties that are especially important for preserving subtle morphological details. The training and sampling procedures of the diffusion model are described in detail in the following subsections.

1) Forward Process: We denote by N the total number of diffusion steps and by $\mathbf{E}_n$ the noisy ECG at step n. The initial ECG segment is $\mathbf{E}_0 \in \mathbb{R}^{S_E}$, where S_E is the number of samples in each ECG segment and $\mathbb{R}^{S_E}$ is the corresponding S_E-dimensional vector space. In the forward process, we construct a sequence $\{\mathbf{E}_n\}_{n=1}^{N}$ by progressively adding Gaussian noise according to a predefined variance schedule $\{\beta_n\}_{n=1}^{N}$:

$$\begin{aligned} &q(\mathbf{E}_n \mid \mathbf{E}_{n-1}) \\ &\quad = \mathcal{N}\big(\mathbf{E}_n; \sqrt{1-\beta_n}\ \mathbf{E}_{n-1}, \beta_n \mathbf{I}\big), \quad n = 1, \ldots, N, \end{aligned} \tag{7}$$

where $q(\cdot)$ is the conditional probability distribution, $\mathcal{N}(\cdot; \upsilon, \zeta)$ represents a multivariate Gaussian distribution with mean υ and covariance ζ, and $\mathbf{I}$ denotes the identity matrix. The forward process is independent of the RCG signals condition, since the added Gaussian noise does not depend on the conditional signal.

The forward process forms a Markov chain that admits a closed-form expression that directly relates $\mathbf{E}_n$ to the initial segment $\mathbf{E}_0$:

$$q(\mathbf{E}_n \mid \mathbf{E}_0) = \mathcal{N}\big(\mathbf{E}_n; \sqrt{\bar{\alpha}_n}\mathbf{E}_0, (1-\bar{\alpha}_n)\mathbf{I}\big), \tag{8}$$

where $\alpha_n = 1 - \beta_n$ and $\bar{\alpha}_n = \prod_{\kappa=1}^{n} \alpha_\kappa$. Therefore, a noisy segment at any timestep n can be obtained as

$$\mathbf{E}_n = \sqrt{\bar{\alpha}_n}\mathbf{E}_0 + \sqrt{1-\bar{\alpha}_n}\varepsilon, \tag{9}$$

where $\varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ denotes a Gaussian noise sample.

2) Reverse Process: Let S_R denote the number of samples in each RCG segment. The reverse diffusion dynamics are represented as another Markov chain that gradually denoises a pure noise segment $\mathbf{E}_N$ into an ECG segment conditioned on $\mathbf{R} \in \mathbb{R}^{S_R}$:

$$\begin{aligned} &p_\theta(\mathbf{E}_{n-1} \mid \mathbf{E}_n, \mathbf{R}) \\ &\quad = \mathcal{N}\big(\mathbf{E}_{n-1}; \mu_\theta(\mathbf{E}_n, n, \mathbf{R}), \sigma_n^2 \mathbf{I}\big), \ n = N, \ldots, 1, \end{aligned} \tag{10}$$

where θ denotes the set of all trainable parameters in the reverse network, $p_\theta(\cdot)$ is the learnable reverse transition distribution parameterized by θ, and σ_n^2 is the variance of the reverse process. Following [38], we set σ_n^2 equal to the forward noise variance, i.e., $\sigma_n^2 = \beta_n$. The mean $\mu_\theta(\mathbf{E}_n, n, \mathbf{R})$ is computed from $\mathbf{E}_n$ and the predicted noise $\varepsilon_\theta(\mathbf{E}_n, n, \mathbf{R})$ as

$$\mu_\theta(\mathbf{E}_n, n, \mathbf{R}) = \frac{1}{\sqrt{\alpha_n}}\left(\mathbf{E}_n - \frac{1-\alpha_n}{\sqrt{1-\bar{\alpha}_n}}\varepsilon_\theta(\mathbf{E}_n, n, \mathbf{R})\right). \tag{11}$$

Consequently, each reverse update can then be expressed as

$$\mathbf{E}_{n-1} = \mu_\theta(\mathbf{E}_n, n, \mathbf{R}) + \sigma_n \varepsilon_r, \tag{12}$$

where $\varepsilon_r \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is the Gaussian noise injected during the reverse process, which we denote separately from the forward process noise ε to avoid confusion.

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To reduce the discrepancy between the estimated noise and the ground-truth noise, we use the mean squared error (MSE) $\mathcal{L}(\theta)$ as the training loss:

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{E}_0,\mathbf{R},\varepsilon,n}[\|\varepsilon - \varepsilon_\theta(\mathbf{E}_n, n, \mathbf{R})\|_2^2]. \tag{13}$$

During the reverse process, we fix the condition $\mathbf{R}$, sample $\mathbf{E}_N$ from a standard normal distribution, and iteratively denoise it according to Eq. (12), yielding a generated ECG $\mathbf{E}^{\text{gen}}$ that is both realistic and aligned with the subject-specific mechanical pattern encoded in the RCG signals.

C. Architecture of the Proposed Conditional Diffusion Model

Under the conditional diffusion framework, the central module is the denoising network $\varepsilon_\theta(\mathbf{E}_n, n, \mathbf{R})$, which predicts the noise contained in a noisy ECG segment $\mathbf{E}_n$ given the diffusion step n and the RCG segment condition $\mathbf{R}$. We implement ε_θ as a 1D residual network with SE channel attention and a temporal self-attention layer, as illustrated in Fig. 3(a). The stacked 1D residual convolutional blocks with SE attention preserve fine-grained local waveform morphology and emphasize diagnostically informative channels, whereas the temporal self-attention layer explicitly models long-range temporal dependencies across multiple beats and R–R intervals. Unlike AirECG [24], which relies on a cross-domain diffusion model built on a hierarchical Transformer with adaptive normalization, our architecture strikes a balance between local detail preservation and global rhythm modeling.

1) Residual Learning with Time and RCG Conditioning: Following the ResNet paradigm, each backbone block is designed to learn a residual mapping rather than a direct transformation. Let C^{res} and S^{res} denote the number of channels and time samples in the feature map, respectively. Given an input feature map $\mathbf{h} \in \mathbb{R}^{C^{\text{res}} \times S^{\text{res}}}$, the residual block computes

$$\mathbf{h}' = \mathbf{h} + F_\varpi(\mathbf{h}), \tag{14}$$

where $\mathbf{h}$ and $\mathbf{h}'$ are the block's input and output feature maps, $F_\varpi(\cdot)$ is the learnable residual mapping implemented by two 1D convolutional layers with non-linear activations and normalization, and ϖ denotes the set of trainable parameters of this residual block.

To make this block aware of the current diffusion step, the scalar index n is first encoded into a time embedding $\mathbf{e}_n$ using sinusoidal positional encoding and then linearly projected to match the channel dimension:

$$\mathbf{h}^{(n)} = \mathbf{h} + \mathbf{W}_n \mathbf{e}_n, \tag{15}$$

where $\mathbf{h}^{(n)}$ is an intermediate representation between $\mathbf{h}$ and $\mathbf{h}'$, and $\mathbf{W}_n$ is a learnable projection matrix for the time embedding. In this way, the response of each block is explicitly modulated by the diffusion step.

Similarly, the convolved RCG segment feature $\mathbf{g}_R$ is incorporated into the block through another linear projection:

$$\mathbf{h}^{(n,R)} = \mathbf{h}^{(n)} + \mathbf{W}_R \mathbf{g}_R, \tag{16}$$

where $\mathbf{h}^{(n,R)}$ is an intermediate representation between $\mathbf{h}^{(n)}$ and $\mathbf{h}'$, and $\mathbf{W}_R$ is a learnable projection matrix for the RCG signal embedding. This operation acts as a FiLM-like conditioning mechanism [39]: the RCG embedding produces channel-wise additive modulation terms that are applied to the ECG features, thereby guiding the denoising process to remain consistent with the mechanical patterns encoded in the RCG.

Accordingly, the input–output relation of the residual block is modified from Eq. (14) to

$$\mathbf{h}' = \mathbf{h} + F_\varpi(\mathbf{h}^{(n,R)}). \tag{17}$$

2) SE Channel Attention: To emphasize diagnostically important channels, we integrate an SE attention module [40] into each residual block. Let C^{se} and S^{se} denote the number of channels and time samples of the SE module input, respectively. Given an intermediate feature map $\mathbf{u} \in \mathbb{R}^{C^{\text{se}} \times S^{\text{se}}}$, where $u_{ch,\ell}$ denotes the value at channel ch and time index ℓ, the SE module first applies a squeeze operation by global average pooling over the temporal dimension:

$$s_{ch} = \frac{1}{S^{\text{se}}} \sum_{\ell=1}^{S^{\text{se}}} u_{ch,\ell}, \; ch = 1, \ldots, C^{\text{se}}, \tag{18}$$

which yields the channel descriptor

$$\mathbf{s} = [s_1, \ldots, s_{C^{\text{se}}}]^\top \in \mathbb{R}^{C^{\text{se}}}. \tag{19}$$

In the excitation stage, $\mathbf{s}$ is fed into a two-layer fully connected bottleneck with reduction ratio τ. Using a SiLU activation [41] followed by a sigmoid, the module produces channel-wise gating coefficients $\mathbf{z} \in \mathbb{R}^{C^{\text{se}}}$:

$$\mathbf{z} = \nu\big(\mathbf{W}_s^2 \; \text{SiLU}(\mathbf{W}_s^1 \mathbf{s})\big), \tag{20}$$

where $\mathbf{W}_s^1 \in \mathbb{R}^{C^{\text{se}}/\tau \times C^{\text{se}}}$ and $\mathbf{W}_s^2 \in \mathbb{R}^{C^{\text{se}} \times C^{\text{se}}/\tau}$ are learnable weight matrices, and $\nu(\cdot)$ is the sigmoid function, which constrains each coefficient to the interval $(0, 1)$. The resulting coefficients are then broadcast along the temporal dimension and applied multiplicatively:

$$u'_{ch,\ell} = z_{ch} \; u_{ch,\ell}, \quad \forall \; ch, \ell, \tag{21}$$

where z_{ch} is the ch-th element of $\mathbf{z}$, producing a reweighted feature map $\mathbf{u}' \in \mathbb{R}^{C^{\text{se}} \times S^{\text{se}}}$. This mechanism allows the network to enhance channels that respond to diagnostically relevant ECG patterns and attenuate less informative ones.

In addition, a self-attention layer is inserted between the encoder and decoder blocks. By combining residual 1D convolutions with SE channel attention and a temporal self-attention layer, the proposed denoising network jointly preserves local ECG morphology and global rhythm structure under RCG signals conditioning. Note that our proposed model is configured to prioritize reconstruction quality rather than strict real-time operation. The inference latency of diffusion model can be reduced by adopting efficient sampling schemes.

D. Arrhythmia Classification Network

The classification stage aims to identify the heartbeat type from the ECG segments generated by the conditional diffusion model. As shown in Fig. 3(b), we design a hybrid CNN-BiLSTM-SE network that integrates local morphology extraction, temporal dependency modeling, and channel-wise attention. The generated ECG segment is first treated as a

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1D temporal sequence and fed into a convolutional front-end composed of several 1D convolution, pooling, and batch normalization blocks. This front-end compresses the features by reducing temporal resolution and channel dimensionality while extracting higher-level local patterns. In this way, the network converts the raw waveform into a compact representation that preserves key morphological information and is more robust to noise and small temporal misalignments.

The compressed convolutional features are then passed through stacked BiLSTM layers, which model temporal dependencies across multiple beats by processing the sequence in both forward and backward directions. This recurrent module enables the network to jointly exploit beat morphology and rhythm-related cues, such as R–R interval variability and inter-beat context. On top of the BiLSTM output, we apply an SE channel attention module that adaptively reweights feature channels based on their global contribution, steering the model toward diagnostically relevant patterns while suppressing redundant responses. The recalibrated sequence is then aggregated by a global average pooling layer and fed into a fully connected classification head with softmax activation to produce posterior probabilities over the heartbeat classes.

V. Experiment Evaluations

A. Dataset and Evaluation Metric

We evaluate the proposed conditional diffusion model based framework using RECG and MIT-BIH Arrhythmia datasets.

- The RECG dataset released by Chen et al. [9] contains synchronized contactless radar cardiography (RCG) and reference ECG recorded with a 77 GHz mmWave radar in a supine setting.
- The MIT-BIH Arrhythmia Dataset [36] consists of 48 half-hour, two-channel Holter ECG recordings from 47 subjects (360 Hz). We extract ECG segments corresponding to the beat types in Fig. 2(b), followed by band-pass filtering and amplitude normalization.

We adopt multiple metrics to comprehensively evaluate the performance of the proposed methods. For ECG reconstruction, we report the validation reconstruction MSE and waveform similarity metrics (Pearson correlation and RMSE), and we evaluate peak-level fidelity by detecting Q/R/S/T events using NeuroKit2 [42]. Following AAMI recommendations[1], a segment is considered correctly monitored if all corresponding peak events deviate by no more than 150 ms; monitoring accuracy is the proportion of correctly monitored segments. Peak localization is further summarized using absolute and relative timing errors. For arrhythmia classification, we report precision, recall, and F1 score.

B. Experimental Setup

We consider two main tasks in our experiments: RCG to ECG conversion and arrhythmia classification. For the conversion task, the conditional diffusion model is trained and evaluated on the RECG dataset, which provides synchronized RCG and ECG signals and thus allows us to quantitatively assess how accurately the model reconstructs ECG waveforms from RCG using the metrics listed in Section V. For the classification task, the CNN-BiLSTM-SE network is trained and evaluated on the MIT-BIH arrhythmia database, whose diverse and well-annotated arrhythmia types enable a rigorous assessment of arrhythmia classification performance on ECG segments. This modular evaluation strategy leverages the complementary strengths of the two datasets. The conditional diffusion model is implemented in PyTorch, whereas the arrhythmia classification network is implemented in TensorFlow. All experimental evaluations are carried out on a workstation featuring an NVIDIA RTX A5000 GPU.

For the conditional diffusion model, we adopt an encoder–decoder architecture with five residual blocks in both the encoder and the decoder, as illustrated in Fig. 3(a). The model has two input channels (the noisy ECG and the RCG condition), and both the batch size and base channel width are set to 64. We employ a 100-step diffusion process with a linear noise schedule, where the variances increase from 1×10^{-4} to 2×10^{-2}. The model is trained for 100 epochs using the Adam optimizer with a learning rate of 1×10^{-4}, minimizing the MSE between the predicted and ground-truth noise. After each epoch, we perform a full reverse diffusion process on a validation batch, starting from pure Gaussian noise and iteratively denoising from x_T to x_0, and record the reconstruction MSE as a validation metric.

The arrhythmia classification network is composed of three CNN layers, two BiLSTM layers, and an SE attention block, as illustrated in Fig. 3(b). It takes a single ECG sequence as input and outputs one of eight heartbeat types. The heartbeat samples are randomly split into training, validation, and test sets in proportions of 64%, 16%, and 20%, respectively. The CNN-BiLSTM-SE network is trained with the Adam optimizer using sparse categorical cross-entropy as the loss function. We train for 50 epochs with a batch size of 64, and report the final performance on the test set after training.

C. ECG Generation Performance on the RECG Dataset

In this section, we assess the ECG generation performance of the proposed conditional diffusion model on the RECG dataset. All paired RCG-ECG segments described in Section V-A are randomly divided into training and test sets in an 80%/20% ratio. The model is trained on the training set and evaluated on the test set. The corresponding quantitative results and waveform analyses are presented in Fig. 4 and Fig. 5.

1) Training Behavior and Monitoring Accuracy: Fig. 4(a) records the evolution of the training loss and validation reconstruction MSE over 100 epochs. We observe that during the first 20 epochs, both curves decrease rapidly, indicating that the diffusion model quickly learns the mapping from RCG to ECG. After this initial phase, both curves gradually converge to lower values. Although the validation MSE exhibits some fluctuations and a few noticeable peaks in the middle of training, this behavior is mainly due to the relatively small validation set and the stochastic nature of optimization. Its overall trend remains consistent with that of the training loss,

[1]https://webstore.ansi.org/standards/aami/ansiaamiec5719982003

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suggesting that the model is properly regularized and does not overfit.

We compare each generated ECG signal with its ground-truth counterpart to determine whether the generation is successful, and then compute the ECG monitoring accuracy for each recording. The results are summarized in Fig. 4(b) and Fig. 4(d). The cumulative distribution function (CDF) in Fig. 4(b) shows that the generated ECG signals achieve high monitoring accuracy across recordings: over 90% of the recordings exceed 95% accuracy, and approximately 20% reach an accuracy greater than 99%. These results suggest that the diffusion model preserves the timing of key ECG events for most segments well enough to satisfy clinical requirements.

Furthermore, we analyze the effect of different physiological conditions on monitoring accuracy, as shown in Fig. 4(d). The CDF curves reveal that monitoring accuracy varies across the four conditions. Specifically, the curves for the post-exercise state are shifted farthest to the left, indicating the lowest monitoring accuracy. This is mainly because intense exercise causes larger chest movements and greater heart rate variability, which make monitoring more challenging. The accuracy curves for sleep and irregular breathing lie between those for post-exercise and normal breathing, indicating that these conditions also increase the difficulty of RCG monitoring and ECG conversion to some extent. Nevertheless, even under irregular breathing, sleep, and post-exercise conditions, 88%, 83%, and 63% of the recordings, respectively, still achieve monitoring accuracy above 95%, demonstrating that our diffusion model remains reliable in challenging physiological conditions.

In a short summary, the convergence curve and the monitoring accuracy distribution indicate that the model learns a robust mapping from RCG to ECG. We then conduct a more detailed comparison between the generated and ground-truth ECG in terms of timing, morphology, and amplitude.

2) Peak Timing Error Analysis: We further assess the timing accuracy of each peak event in the generated ECG. The R–R interval between successive ECG beats reflects the duration of a single cardiac cycle and allows estimation of the heart rate, which is clinically important. Fig. 4(c) illustrates the CDF of the absolute R-R interval error across all generated ECG segments. From the figure, we can see that more than 65% of R-R intervals are generated accurately, while fewer than 3% have an absolute timing error greater than 10 ms. These results indicate that the generated ECG signals exhibit very little timing drift in the R-R intervals relative to the ground-truth ECG.

In Fig. 4(e), we present the relative timing errors of the peak events under different physiological conditions. The third quartile (Q3) is used as an indicator of relative timing error: Q3 means that 75% of the relative timing errors for a given generated peak are smaller than this value. For normal breathing, the Q3 values for the Q, R, S, and T peaks are 0.62%, 0%, 0.6%, and 0.57%, respectively. For sleep, the corresponding Q3 values are 0.69%, 0%, 0.51%, and 0.65%. For irregular breathing, they are 0.61%, 0%, 0.59%, and 0.59%. And for post-exercise, the Q3 values are 0.74%, 0%, 0.67%, and 0.72%. Overall, the relative timing errors for all peaks are small, suggesting that our model can reliably generate key ECG events. On the other hand, in the post-exercise state, substantial chest motion leads to slightly larger timing errors for each peak compared with the other physiological conditions, which is consistent with the monitoring accuracy analysis in the previous section.

We investigate the effect of physiological state on long-term ECG conversion with the proposed conditional diffusion model. Fig. 4(f) plots the CDF of the absolute R-R interval timing errors under different physiological conditions. All four curves rise sharply at small error values and are very similar to one another. This indicates that the absolute R–R interval timing error of the generated ECG signals shows only weak dependence on physiological state.

In a short summary, the proposed conditional diffusion model performs well in both instantaneous peak event localizations and beat-to-beat timing in ECG conversion (R-R intervals), indicating that it can accurately capture the temporal relationships of key ECG events when converting from RCG to ECG.

3) Morphology Analysis: To visually compare the generated ECG with the ground-truth signal, we select representative signal pairs from the four physiological conditions and overlay each pair in the same panel, as shown in Figs. 5(a), (b), (e), and (f). For each pair, we also compute the Pearson correlation coefficient and the amplitude RMSE, and report these values at the top of the figure. Figs. 5(a) and (e) illustrate cases where the generated waveform closely matches the ground-truth waveform, characterized by a high correlation coefficient and low amplitude RMSE. In Fig. 5(b), the generated ECG waveform is globally shifted relative to the ground-truth, which lowers the correlation coefficient and increases the amplitude RMSE. However, it still reflects the true ECG to some extent because its R–R intervals remain aligned with those of the ground-truth waveform. Fig. 5(f) shows a more severe failure case in which part of the generated ECG is shifted, resulting in a low correlation coefficient and a high amplitude RMSE, and the generated waveform no longer reflects the true ECG.

In a short summary, these examples indicate that the correlation coefficient and amplitude RMSE are effective indicators for assessing whether an RCG to ECG conversion is successful. Therefore, in the next step, we use these two metrics to analyze all segments across all recordings.

4) Amplitude and Correlation Analysis: In Fig. 5(c), we show the RMS amplitude errors of the peak events under different physiological conditions. For normal breathing, the median RMSE values for the Q, R, S, and T peaks are 0.006, 0.016, 0.008, and 0.009 mV, respectively. For sleep, the corresponding median RMSE values are 0.007, 0.015, 0.015, and 0.009 mV. For irregular breathing, they are 0.006, 0.016, 0.009, and 0.008 mV. And for post-exercise, the median RMSE values are 0.006, 0.017, 0.012, and 0.010 mV. Overall, the median RMSE for each peak is small, and the interquartile range is relatively narrow, indicating that the model effectively preserves the amplitude of the ECG waveform peaks. Specifically, the Q peak has the lowest RMSE, while the R peak has the highest. This is likely because the R peak is the sharpest

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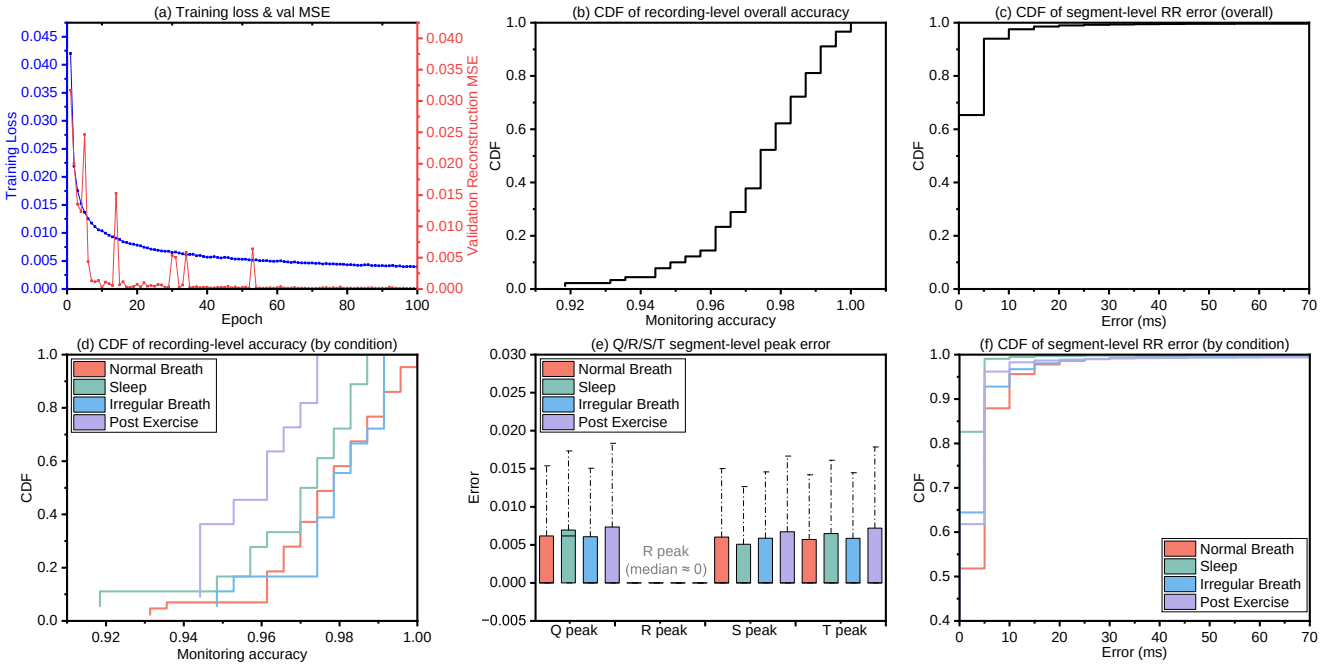

Fig. 4. Overall accuracy and timing error performance of the proposed contactless heartbeat monitoring system (each recording consists of multiple segments). (a) Training loss and validation reconstruction MSE over epochs, showing stable convergence. (b) CDF of recording-level monitoring accuracy, with most recordings achieving high accuracy. (c) CDF of overall segment-level R-R interval timing errors, indicating small deviations. (d) CDF of recording-level monitoring accuracy under different breathing and exercise conditions. (e) Segment-level Q/R/S/T peak timing errors for each condition. (f) CDF of segment-level R-R interval timing errors by condition, demonstrating robust timing across scenarios.

among the four peak types, and even small errors or distortions during conversion can greatly impact its RMSE. In contrast, the Q peak is less affected by such issues. Additionally, the lower RMSE for the T peak is likely attributed to its broader and smoother shape, which makes it more resistant to noise and distortion.

We also assess the linear similarity between the generated and ground-truth ECG waveforms using the Pearson correlation coefficient. Fig. 5(d) reports the distribution of correlation coefficients under different physiological conditions. The median correlation values for normal breathing, sleep, irregular breathing, and post-exercise conditions are 99.75%, 99.74%, 99.71%, and 99.70%, respectively, while the correlation Q3 values are 99.83%, 99.81%, 99.82%, and 99.79%. The medians of all four box plots are very high and the interquartile ranges are very small, confirming that the generated ECG signals closely match the reference ECG in overall morphology. The median and Q3 correlation coefficients for the post-exercise conditions are slightly lower, which is consistent with the earlier analysis.

Finally, Fig. 5(g) and Fig. 5(h) present the CDFs of the RMS amplitude error and the correlation coefficient across all segments, respectively. The CDF of RMSE rises sharply near zero. 90% of segments have an RMSE below 0.024 mV, and fewer than 1% exceed 0.1 mV, indicating that most generated segments exhibit only minor amplitude deviations from the ground-truth. Conversely, the CDF of the correlation coefficient remains close to zero for low correlation values and rises sharply near one. Fewer than 1% of segments have correlation coefficients below 95.84%, while more than 93.4% exceed 99%, showing that most generated segments are highly correlated with the ground-truth ECG waveforms.

In a short summary, the proposed conditional diffusion model can reproduce ECG amplitude and overall waveform shape with high fidelity. At this stage, we have verified the effectiveness of the model for RCG to ECG conversion from four perspectives: monitoring accuracy, key event timing accuracy, peak amplitude, and overall waveform morphology.

D. Leave-One-Person-Out Evaluation on the RECG Dataset

In real contactless monitoring scenarios, the model may generalize to new users whose data are not seen during training. To evaluate the subject-independent generalization ability of the proposed method, we conduct a leave-one-person-out (LOPO) experiment [43] on the RECG dataset. In the LOPO setting, each chosen subject–condition pair is treated as an independent test case. For a given pair, all segments from that subject under a specific physiological condition are used only for testing, while segments from all other subjects are used to train the conditional diffusion model. The training procedure and hyperparameters are kept identical to those in the previous experiments. This process is repeated for all selected pairs and the test results are aggregated. The results are summarized in Fig. 6.

Leave-one-person-out (LOPO) cross-validation results of the proposed contactless heartbeat monitoring system under four conditions. (a) Recording-level monitoring accuracy by condition, showing reduced accuracy in more challenging post-exercise scenarios. (b) Segment-level Q/R/S/T peak timing errors by condition. (c) Segment-level Q/R/S/T peak RMS amplitude errors by condition, with larger errors under irregular breathing and post-exercise.

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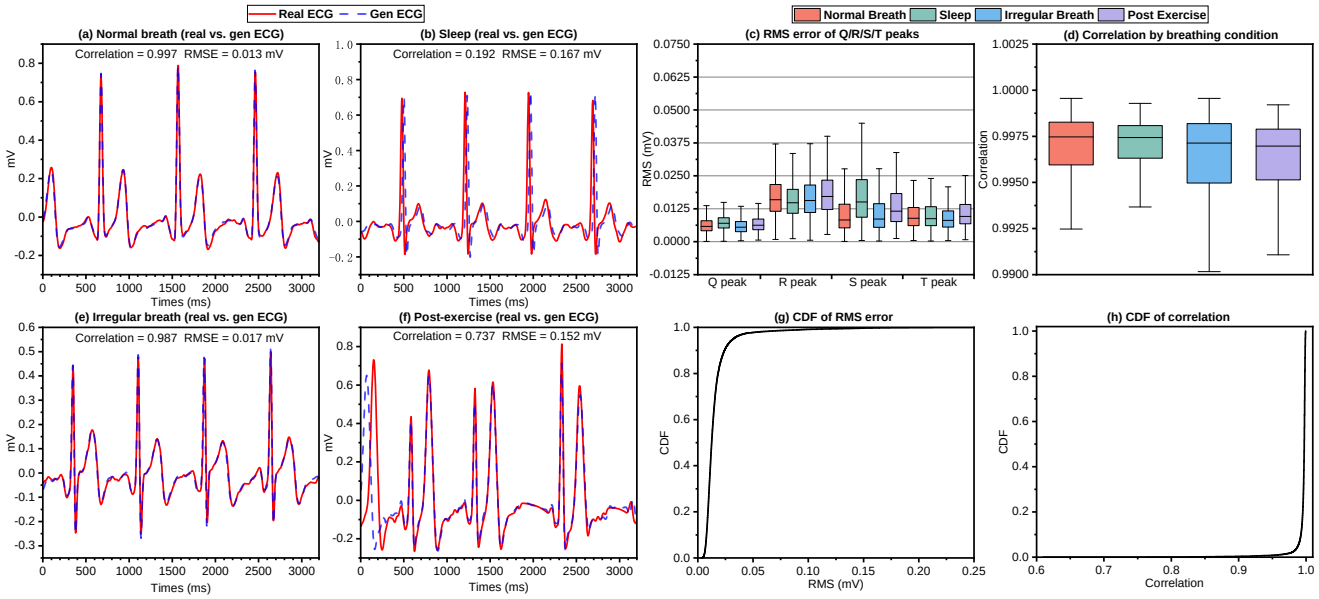

Fig. 5. ECG morphology examples and segment-level reconstruction accuracy of the proposed contactless heartbeat monitoring system. (a), (b), (e), and (f) show real vs. generated ECG segments under four conditions, illustrating close alignment in waveform morphology and amplitude. (c) RMSE of Q/R/S/T peak amplitudes by condition. (d) Correlation between real and generated ECG by condition, remaining high across all conditions. (g) CDF of RMS amplitude error across all segments, with most segments exhibiting very low reconstruction error. (h) CDF of Pearson correlation across all segments, with the majority achieving high correlation values.

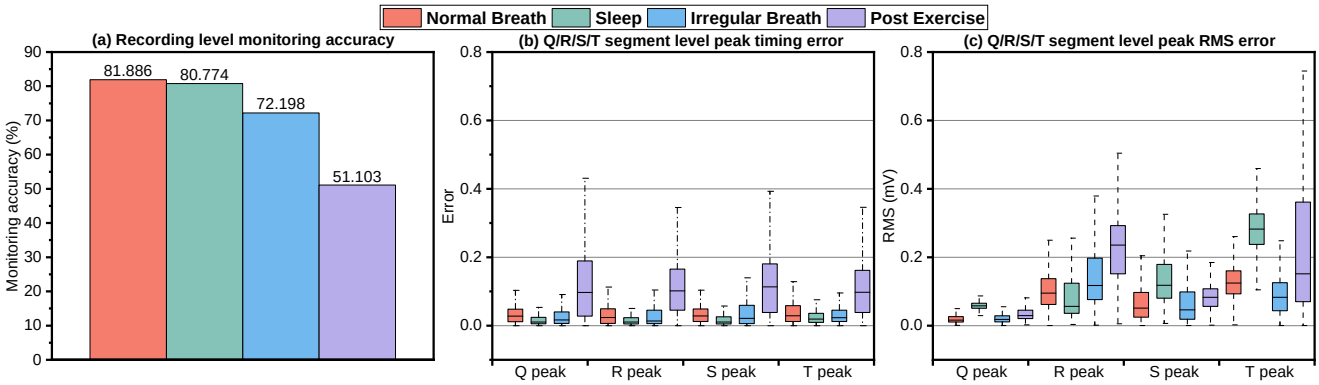

Fig. 6. Leave-one-person-out (LOPO) cross-validation results of the proposed contactless heartbeat monitoring system under four breathing conditions. (a) Recording-level monitoring accuracy by condition, showing reduced accuracy in more challenging scenarios. (b) Segment-level Q/R/S/T peak timing errors by condition. (c) Segment-level Q/R/S/T peak RMS amplitude errors by condition, with larger errors under post-exercise.

1) Monitoring Accuracy Under Different Conditions: Fig. 6(a) summarizes the LOPO monitoring accuracies under the four physiological conditions. Compared with the experiment in Section V-C, the monitoring accuracies for all conditions are noticeably lower. This is expected, since the LOPO protocol enforces strict subject independence, making it a more challenging scenario. Specifically, the monitoring accuracies for normal breathing, sleep, irregular breathing, and post-exercise conditions are 81.89%, 80.77%, 72.20%, and 51.10%, respectively. Overall, the proposed conditional diffusion model is still capable of accurately monitoring most segments from unseen subjects, especially in the relatively stable conditions of normal breathing and sleep. For irregular breathing and post-exercise conditions, higher heart rate variability and more complex chest motion make RCG to ECG conversion more difficult.

2) Peak Timing Error Under LOPO Setting: We further examine the impact of the LOPO protocol on the timing accuracy of individual peak events. Fig. 6(b) shows box plots of the relative timing errors for the peak events under the four physiological conditions. For normal breathing, the median relative timing errors for the Q, R, S, and T peaks are 2.76%, 2.37%, 2.83%, and 2.86%, respectively. For sleep, the corresponding medians are 1.07%, 1.03%, 1.03%, and 1.92%. For irregular breathing, they are 1.68%, 1.34%, 2.12%, and 2.33%. And for post-exercise, the median relative timing errors are 9.72%, 10.15%, 11.32%, and 9.77%. Overall, the relative timing errors for all peak types increase under LOPO. However, for normal breathing, sleep, and irregular breathing, the medians remain small and the interquartile ranges are narrow, indicating that for unseen subjects most peaks still lie close to their true positions. As for the post-exercise condition, the behavior is consistent with previous experiments: the

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median relative timing error is higher and the interquartile range is wider, but the timing deviations are still within an acceptable range.

3) RMS Amplitude Error Under LOPO Setting: We also investigate the influence of the LOPO protocol on the amplitude fidelity of the generated ECG. Fig. 6(c) illustrates the RMS amplitude errors of the peak events under the four physiological conditions. For normal breathing, the median RMSE values for the Q, R, S, and T peaks are 0.016, 0.095, 0.051, and 0.124 mV, respectively. For sleep, the corresponding medians are 0.058, 0.056, 0.118, and 0.282 mV. For irregular breathing, they are 0.018, 0.117, 0.046, and 0.083 mV. And for post-exercise, the median RMSE values are 0.029, 0.236, 0.083, and 0.152 mV. Across all conditions, the Q peak still has the lowest RMSE, whereas the R peak maintains the highest RMSE. Moreover, we observe a clear increase in RMSE for all peaks during sleep, especially for the T wave. This may be because the T wave has a broad shape and exhibits large inter-individual variation. In addition, subjects may change posture during sleep, which can weaken the radar echoes. These factors collectively lead to higher RMSE values for all peaks during sleep, with the T wave being most affected. Nevertheless, for all physiological conditions, most RMSE values remain well below the upper limit of the plots, indicating that under the LOPO setting, the generated ECG still reproduces the peak events amplitudes with acceptable accuracy.

In a short summary, the LOPO experiment shows that the proposed conditional diffusion model maintains reasonable performance when applied to unseen subjects, although monitoring accuracy as well as peak timing and amplitude accuracy all degrade compared with the results in Section V-C. These findings indicate that the mapping from RCG to ECG does not solely rely on subject-specific characteristics. The proposed framework exhibits a certain level of subject-independent generalization, which is crucial for practical contactless monitoring applications.

E. Arrhythmia Classification Performance on MIT-BIH

In our framework, the ECG signals generated from RCG are ultimately fed into the CNN-BiLSTM-SE network for arrhythmia classification. To verify that this network can effectively capture diagnostically relevant ECG patterns and to quantify the contribution of each component, we conduct an ablation study on the MIT-BIH Arrhythmia Database. Specifically, we train the proposed model and all ablated variants using the same training protocol and data split as described in Section IV-D, which ensures that any performance differences arise from architectural changes rather than training conditions. We then evaluate all trained networks on the test set and record their performance, as summarized in Fig. 7.

1) Comparison with Ablation Models: Figs. 7(a)-(c) compare the per-class precision, recall, and F1 scores of the proposed CNN-BiLSTM-SE network with its ablated variants (CNN, LSTM, BiLSTM, CNN-SE, LSTM-SE, and BiLSTM-SE). Overall, the proposed model achieves the highest or near-highest precision, recall, and F1 scores across all eight heartbeat types, showing the advantage of combining convolutional feature extraction, bidirectional temporal modeling, and channel-wise attention within a single architecture. Specifically, for heartbeat types with abundant samples (N, /, L, R, and V), all models perform relatively well. In contrast, for types with scarce samples (A, F, and f), test performance differs markedly across models. For these rare categories, using only CNN, LSTM, or BiLSTM leads to a clear performance drop. Adding the SE attention mechanism to these models helps mitigate this issue, but the best performance is obtained only when CNN, BiLSTM, and SE attention are integrated together. These results indicate that convolution alone cannot capture long-range temporal dependencies, and recurrent layers without prior convolutional preprocessing struggle to model detailed waveform shapes. By contrast, the components in our proposed network complement one another, and their combination is crucial for accurately identifying challenging heartbeat types with limited samples.

2) Per-Class Performance and Confusion Matrix: To further analyze the behavior of the proposed classifier, Fig. 7(c) shows the confusion matrix of the CNN-BiLSTM-SE network on the MIT-BIH test set. The matrix is heavily concentrated along the main diagonal, indicating that most heartbeats are correctly classified and that the overall recognition performance is high. We then focus on several rhythm types that are frequently misclassified. Atrial premature beats (A) are sometimes predicted as normal heartbeats, which is consistent with their relatively subtle deviation from normal morphology. Premature ventricular contractions (V) and fusion beats (F) are often confused with each other and are occasionally misclassified as normal beats, reflecting the similarity of their ventricular QRS patterns.

In summary, the ablation study and confusion matrix show that the proposed CNN-BiLSTM-SE classifier delivers highly reliable predictions for the dominant heartbeat types while still maintaining reasonable discriminative ability for less frequent and more challenging arrhythmias.

VI. Conclusion

In this paper, we have proposed a contactless ECG monitoring method that integrates mmWave radar sensing and generative AI. RCG signals are first acquired using mmWave radar and then input to a ResNet-based conditional diffusion model to synthesize the corresponding ECG waveforms. Finally, a CNN-BiLSTM-SE network is employed to perform arrhythmia classification. Experiments on the RECG dataset have shown that the ECG signals generated by our model are effective in terms of monitoring accuracy, key event timing, peak amplitude, and overall waveform morphology. In addition, we have demonstrated that the diffusion model provides a reasonable degree of subject-independent generalization. Moreover, experiments on the MIT-BIH dataset have confirmed that the CNN-BiLSTM-SE classifier performs well for both common and rare heartbeat types, making it suitable for practical arrhythmia classification.

Despite these promising results, several limitations remain. First, the current system provides reliable monitoring only

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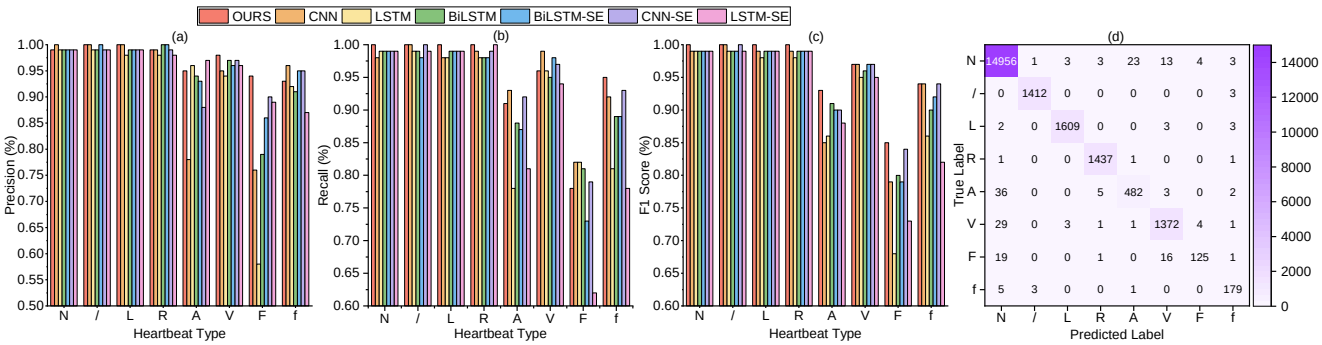

Fig. 7. Overall arrhythmia classification performance of the proposed model and ablation baselines. **(a)**: Precision by heartbeat type for different models. **(b)**: Recall by heartbeat type for different models. **(c)**: F1 scores by heartbeat type for different models, highlighting the advantages of the proposed model, especially for rare classes. **(d)**: Confusion matrix of heartbeat classification for the proposed model, with most samples correctly classified.

when the subject is at rest and cannot robustly acquire data during body movement. Second, the data analyzed in this study were collected in relatively low-noise environments. Monitoring in crowded or office settings will be more challenging due to increased interference and multipath effects. Consequently, in future work we plan to extend the proposed framework to support reliable monitoring of moving subjects in more complex and realistic settings.

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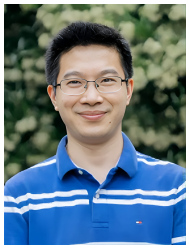
Hanwen Zhang received the B.Sc. degree from the University of Electronic Science and Technology of China, China, in 2023. He is currently pursuing the Ph.D. degree with the Department of Computer and Information Science, University of Macau, Macau, China. His research interests include edge computing, millimeter-wave radar, and deep learning.

Peichun Li received his M.Sc. degree from the Guangdong University of Technology, China, in 2021. He is currently pursuing the Ph.D. degree with the Department of Computer and Information Science, University of Macau, Macau, China. His research interests include edge computing and deep learning, particularly in efficient algorithms for artificial intelligence (AI) applications.

Liping Qian (Senior Member, IEEE) received the Ph.D. degree in information engineering from The Chinese University of Hong Kong in 2010. She is currently a Full Professor with the Institute of Cyberspace Security, Zhejiang University of Technology, Hangzhou, China, and an adjunct professor with the School of Ocean Engineering and Technology, Sun Yat-Sen University, Zhuhai, China. Her research interests include wireless communication and networking, resource management in wireless networks, massive IoTs, mobile edge computing, emerging multiple access techniques, and machine learning oriented towards wireless communications. She was a co-recipient of the IEEE Marconi Prize Paper Award in Wireless Communications in 2011 and the Best Paper Awards from IEEE ICC in 2016, IEEE Communication Society GCCTC in 2017, the Digital Communications and Networking in 2021, and the 2023 IEEE WCNC. She is the Distinguished Lecturer of the IEEE Vehicular Technology Society from 2024 to 2026. She is on the editorial board of IEEE Transactions on Cognitive Communications and Networking, IEEE Internet of Things Journal, and IEEE Wireless Communications.



Zhiguo Shi (Fellow, IEEE) received the B.S. and Ph.D. degrees in electronic engineering from Zhejiang University, Hangzhou, China, in 2001 and 2006, respectively. Since 2006, he has been a Faculty Member with the College of Information Science and Electronic Engineering, Zhejiang University, where he is currently a Full Professor. From 2011 to 2013, he visited the Broadband Communications Research Group, University of Waterloo, Waterloo, ON, Canada. His research interests include array signal processing, localization, and the Internet of Things with applications to drone surveillance. He was the General Co-Chair of IEEE SAM 2020 and served as an Editor of IEEE Network and IEEE Transactions on Circuits and Systems—I: Regular Papers. He is serving as an Area Editor for IEEE Internet of Things Journal and an Associate Editor for IEEE Signal Processing Letters, IEEE Transactions on Vehicular Technology, and IEEE Transactions on Network Science and Engineering.

Yuan Wu (Senior Member, IEEE) is currently a Full Professor with the State Key Laboratory of Internet of Things for Smart City and the Department of Computer and Information Science, University of Macau, Macau, SAR, China. His research interests include resource management for wireless networks, mobile edge computing and edge intelligence, and integrated sensing and communications.

Dr. Wu was a recipient of the Best Paper Award from the IEEE ICC 2016, IEEE TCGCC 2017, IWCMC 2021, and IEEE WCNC 2023. He serves on the Editorial Board of IEEE Transactions on Wireless Communications, IEEE Transactions on Vehicular Technology, and IEEE Transactions on Network Science and Engineering. He is a Distinguished Lecturer of the IEEE Vehicular Technology Society (VTS).

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